

SIEMENS EDA

Algorithmic C (AC) Floating-Point Math Library Release Notes

Software Version v3.8.1
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Release 3.8.1

Enhanced the supported features of the `ac_block_float` type.

The `ac_block_float` datatype was enhanced by the following features:

- Enhanced to support subnormal values.
- Enhanced to meters.

Introduced a new dual-path architecture for the standard datatypes in `ac_std_float.h` (namely `ac_std_float`, `ac_ieee_float` and `bfloat16`).

Added a new set of functions that implement a dual-path architecture for standard floating-point addition. The newly introduced functions support the standard floating-point datatypes of `ac_std_float.h` (`ac_std_float`, `ac_ieee_float` and `bfloat16`). These functions are supposed to generate designs with reduced delay, compared to the standard addition of `ac_std_float`, by increasing the area. They are developed for cases where the aim is to improve the latency of the design.

Release 3.6.0

This version of `ac_fpmath` was included for the first time in Catapult release 2025.1.

Added Block Floating-Point

Added the block floating-point datatype that offers a low delay solution for floating point additions, ideal for designing accumulators.

Added Standard Floating-Point Dot Product Function

Added the `fdot_tree()` and `fdot_tree_ptr()` functions that implements dot product operation for an arbitrary number of inputs. The function support `ac_std_float`, `ac_ieee_float` and `ac::bfloat16` datatypes.

Added Standard Floating-Point Square Function

Added the `ac_sqr()` function that implements a square operation a^2 for the `ac_std_float` datatype.